

- Surowiecki, J. (2004). *The Wisdom of Crowds: Why the Many are Smarter than the Few and How Collective Wisdom Shapes Business, Economics, Societies and Nations.*, Little, Brown.
- Swayne, D., Cook, D. and Buja, A. (1991). Xgobi: Interactive dynamic graphics in the X window system with a link to S, *ASA Proceedings of Section on Statistical Graphics*, pp. 1–8.
- Tanner, M. and Wong, W. (1987). The calculation of posterior distributions by data augmentation (with discussion), *Journal of the American Statistical Association* **82**: 528–550.
- Tarpey, T. and Flury, B. (1996). Self-consistency: A fundamental concept in statistics, *Statistical Science* **11**: 229–243.
- Tenenbaum, J. B., de Silva, V. and Langford, J. C. (2000). A global geometric framework for nonlinear dimensionality reduction, *Science* **290**: 2319–2323.
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso, *Journal of the Royal Statistical Society, Series B* **58**: 267–288.
- Tibshirani, R. and Hastie, T. (2007). Margin trees for high-dimensional classification, *Journal of Machine Learning Research* **8**: 637–652.
- Tibshirani, R. and Knight, K. (1999). Model search and inference by bootstrap “bumping”, *Journal of Computational and Graphical Statistics* **8**: 671–686.
- Tibshirani, R. and Wang, P. (2007). Spatial smoothing and hot spot detection for CGH data using the fused lasso, *Biostatistics* **9**: 18–29.
- Tibshirani, R., Hastie, T., Narasimhan, B. and Chu, G. (2001a). Diagnosis of multiple cancer types by shrunken centroids of gene expression, *Proceedings of the National Academy of Sciences* **99**: 6567–6572.
- Tibshirani, R., Hastie, T., Narasimhan, B. and Chu, G. (2003). Class prediction by nearest shrunken centroids, with applications to DNA microarrays, *Statistical Science* **18**(1): 104–117.
- Tibshirani, R., Saunders, M., Rosset, S., Zhu, J. and Knight, K. (2005). Sparsity and smoothness via the fused lasso, *Journal of the Royal Statistical Society, Series B* **67**: 91–108.
- Tibshirani, R., Walther, G. and Hastie, T. (2001b). Estimating the number of clusters in a dataset via the gap statistic, *Journal of the Royal Statistical Society, Series B.* **32**(2): 411–423.
- Tropp, J. (2004). Greed is good: algorithmic results for sparse approximation, *IEEE Transactions on Information Theory* **50**: 2231–2242.

- Tropp, J. (2006). Just relax: convex programming methods for identifying sparse signals in noise, *IEEE Transactions on Information Theory* **52**: 1030–1051.
- Valiant, L. G. (1984). A theory of the learnable, *Communications of the ACM* **27**: 1134–1142.
- van der Merwe, A. and Zidek, J. (1980). Multivariate regression analysis and canonical variates, *The Canadian Journal of Statistics* **8**: 27–39.
- Vapnik, V. (1996). *The Nature of Statistical Learning Theory*, Springer, New York.
- Vapnik, V. (1998). *Statistical Learning Theory*, Wiley, New York.
- Vidakovic, B. (1999). *Statistical Modeling by Wavelets*, Wiley, New York.
- von Luxburg, U. (2007). A tutorial on spectral clustering, *Statistics and Computing* **17**(4): 395–416.
- Wahba, G. (1980). Spline bases, regularization, and generalized cross-validation for solving approximation problems with large quantities of noisy data, *Proceedings of the International Conference on Approximation theory in Honour of George Lorenz*, Academic Press, Austin, Texas, pp. 905–912.
- Wahba, G. (1990). *Spline Models for Observational Data*, SIAM, Philadelphia.
- Wahba, G., Lin, Y. and Zhang, H. (2000). GACV for support vector machines, in A. Smola, P. Bartlett, B. Schölkopf and D. Schuurmans (eds), *Advances in Large Margin Classifiers*, MIT Press, Cambridge, MA., pp. 297–311.
- Wainwright, M. (2006). Sharp thresholds for noisy and high-dimensional recovery of sparsity using ℓ_1 -constrained quadratic programming, *Technical report*, Department of Statistics, University of California, Berkeley.
- Wainwright, M. J., Ravikumar, P. and Lafferty, J. D. (2007). High-dimensional graphical model selection using ℓ_1 -regularized logistic regression, in B. Schölkopf, J. Platt and T. Hoffman (eds), *Advances in Neural Information Processing Systems 19*, MIT Press, Cambridge, MA, pp. 1465–1472.
- Wasserman, L. (2004). *All of Statistics: a Concise Course in Statistical Inference*, Springer, New York.
- Weisberg, S. (1980). *Applied Linear Regression*, Wiley, New York.

- Werbos, P. (1974). *Beyond Regression*, PhD thesis, Harvard University.
- Weston, J. and Watkins, C. (1999). Multiclass support vector machines, in M. Verleysen (ed.), *Proceedings of ESANN99*, D. Facto Press, Brussels.
- Whittaker, J. (1990). *Graphical Models in Applied Multivariate Statistics*, Wiley, Chichester.
- Wickerhauser, M. (1994). *Adapted Wavelet Analysis from Theory to Software*, A.K. Peters Ltd, Natick, MA.
- Widrow, B. and Hoff, M. (1960). Adaptive switching circuits, *IRE WESCON Convention record*, Vol. 4. pp 96-104; Reprinted in Andersen and Rosenfeld (1988).
- Wold, H. (1975). Soft modelling by latent variables: the nonlinear iterative partial least squares (NIPALS) approach, *Perspectives in Probability and Statistics, In Honor of M. S. Bartlett*, pp. 117–144.
- Wolpert, D. (1992). Stacked generalization, *Neural Networks* **5**: 241–259.
- Wu, T. and Lange, K. (2007). The MM alternative to EM, unpublished.
- Wu, T. and Lange, K. (2008). Coordinate descent procedures for lasso penalized regression, *Annals of Applied Statistics* **2**(1): 224–244.
- Yee, T. and Wild, C. (1996). Vector generalized additive models, *Journal of the Royal Statistical Society, Series B.* **58**: 481–493.
- Yuan, M. and Lin, Y. (2007). Model selection and estimation in regression with grouped variables, *Journal of the Royal Statistical Society, Series B* **68**(1): 49–67.
- Zhang, P. (1993). Model selection via multifold cross-validation, *Annals of Statistics* **21**: 299–311.
- Zhang, T. and Yu, B. (2005). Boosting with early stopping: convergence and consistency, *Annals of Statistics* **33**: 1538–1579.
- Zhao, P. and Yu, B. (2006). On model selection consistency of lasso, *Journal of Machine Learning Research* **7**: 2541–2563.
- Zhao, P., Rocha, G. and Yu, B. (2008). The composite absolute penalties for grouped and hierarchical variable selection, *Annals of Statistics*. (to appear).
- Zhu, J. and Hastie, T. (2004). Classification of gene microarrays by penalized logistic regression, *Biostatistics* **5**(2): 427–443.
- Zhu, J., Zou, H., Rosset, S. and Hastie, T. (2005). Multiclass adaboost, Unpublished.

- Zou, H. (2006). The adaptive lasso and its oracle properties, *Journal of the American Statistical Association* **101**: 1418–1429.
- Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net, *Journal of the Royal Statistical Society Series B.* **67**(2): 301–320.
- Zou, H., Hastie, T. and Tibshirani, R. (2006). Sparse principal component analysis, *Journal of Computational and Graphical Statistics* **15**(2): 265–28.
- Zou, H., Hastie, T. and Tibshirani, R. (2007). On the degrees of freedom of the lasso, *Annals of Statistics* **35**(5): 2173–2192.

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